



NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

Software-defined EV chips

ST Microelectronics Stellar E'MCU microcontrollers are specifically for software-defined electric vehicles (EVs). The E'MCU family includes the Stellar P series for integration and vehicle control and the Stellar G series for body applications. These MCUs are dedicated to facilitating centralised (domain and zonal) electronics architectures and simplifying high-efficiency power modules for e-drivetrain charging and digital power conversion. The MCUs can easily be upgraded in the future through secure over-the-air (OTA) updates.

ST Microelectronics
<https://www.st.com>

12th Gen mobile processors

The 12th gen Intel core HX processors are designed to provide high-level performance of mobile workstation platforms. These include seven new



mobile processors available in the existing configuration of Core i5, Core i7, and Core i9 models.

The desktop-caliber mobile processors have eight performance cores and eight efficient cores with 24 threads running at a base power of 55W.

These are suitable for power-intensive tasks, such as gaming and professional workflows like visual effects, CAD, and video rendering.

Intel
<https://www.intel.com>

3kA surge-protection thyristors

Littelfuse Pxxx0S3N SIDAcTtor Protection thyristor series safeguard exposed interfaces in industrial and ICT

applications, including RS-485 data interfaces used in factory automation and AC/DC power supplies used in energy storage systems and electric vehicle wall chargers. The thyristors provide 3kA (8/20) high-power surge protection with low thermal accumulation during long-term overvoltage events, so can be incorporated in equipment used in harsh environments.



Littelfuse
<https://www.littelfuse.com>

Radiation-tolerant flash

Microchip's SST26LF064RT is a 64-Mbit serial quad I/O NOR flash memory device for use in harsh aerospace and defense system environments. The commercial off-the-shelf memory device



has high radiation tolerance and is capable of withstanding 50-kilorad total ionizing dose (TID). The space-qualified second SuperFlash device from Microchip reduces development time, cost, and risk in the systems. The IC also reduces the complexity of power management switching to achieve the high TID even while the flash is still biased and operating in systems such as satellite onboard computers.

Microchip
<https://www.microchip.com>

IPM for 3-phase inverters

CIPOS Tiny IM323-L6G is a new addition to Infineon Technologies' intelligent power module (IPM) series. The IPM module is optimised for 3-phase

inverters with an operating range of 1 to 20kHz. It is rated for 600V, 15A and is based on the TRENCHSTOP IGNT



RC-D2 switches and the advanced SOI gate drivers. The IM323-L6G

offers high efficiency and better reliability even with lower size and cost. It protects from short circuits, over-current, and under-voltage.

Infineon Technologies
<https://www.infineon.com>

Fast-recovery MOSFETs

The R60xxVNX series of MOSFETs provide a low on-resistance along with one of the fastest reverse recovery times. The series is optimised for power circuits in industrial equipment requiring high power, such as servers, EV charging, motor drives, etc. According to the company, the move towards a carbon-neutral world has forced us to become more energy-efficient, demanding power semiconductors to further reduce power loss. To meet the expectations, the R60xxVNX series uses the latest proprietary processes to achieve the industry's fastest reverse recovery time while reducing on resistance (which is in a trade-off relationship) by up to 20% compared to equivalent products.

ROHM Semiconductor
<https://www.rohm.com>

Solid-state EV relays

Texas Instruments TPSI3050-Q1 is an isolated switch driver with an integrated 10V gate supply and the TPSI2140-Q1, a 1400V, 50 mA isolated switch. The new series has a smaller size and lower power consumption along with a lower cost, making them suitable for use in electric vehicles (EVs). These solid-state relays are capable of achieving higher system